

Topic 12: Getting Started with Generative AI

12.1 Tools to Sign Up for Today

- ChatGPT (chat.openai.com): free tier uses GPT-4o mini; Plus (\$20/month) gives GPT-4o with image input and generation
- Claude (claude.ai): free tier available; excellent for long documents, nuanced writing, and complex instruction-following
- Google Gemini (gemini.google.com): free with a Google account; integrates with Gmail, Docs, and Drive
- Microsoft Copilot (copilot.microsoft.com): free, powered by GPT-4; integrated into Windows and Edge
- Perplexity AI (perplexity.ai): AI-powered search with cited sources; free tier available -- great for research
- GitHub Copilot (github.com/features/copilot): AI coding assistant; free for students and open-source contributors

12.2 Your First Steps

- Use ChatGPT or Claude every day for tasks you currently do manually: emails, summaries, research questions, code help
- Learn prompt engineering: be specific about your goal, audience, format, and tone -- vague prompts produce vague output
- Get API access (openai.com/api or console.anthropic.com) and make your first API call in Python
- Build a small project: a document summariser, a chatbot for a FAQ, a code reviewer -- learning by building is fastest
- Explore Ollama (ollama.com) to run open-source models like Llama 3 locally, free, with full privacy

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# Your first OpenAI API call (Python):
pip install openai

from openai import OpenAI
client = OpenAI(api_key='your-key-here')

response = client.chat.completions.create(
    model='gpt-4o-mini',
    messages=[{'role': 'user', 'content': 'Explain neural networks in 3 sentences.'}]
)
print(response.choices[0].message.content)
```

12.3 Recommended Learning Path

- Beginner (this course): understand what generative AI is, how it works, and the key tools
- Next -- Prompt Engineering: learn to write effective prompts that consistently produce high-quality output
- Next -- AI Tools for Everyday Use: master ChatGPT, Claude, Copilot, Midjourney, Notion AI in depth
- Intermediate -- Working with LLM APIs: build applications in Python using the OpenAI, Anthropic, or Gemini API
- Intermediate -- RAG (Retrieval-Augmented Generation): connect AI to your own documents and databases
- Intermediate -- AI Agents: build AI systems that use tools, browse the web, and complete multi-step tasks
- Advanced: Fine-Tuning, Evaluation, Production AI, MLOps

12.4 Staying Current

- AI capabilities change faster than almost any other technology -- stay curious and experiment regularly
 - Newsletters: The Batch (Andrew Ng, deeplearning.ai), TLDR AI, The Neuron, Lenny's Newsletter
 - Research papers: arxiv.org/cs.AI -- read abstracts of key papers; skim full papers for landmark work
 - Community: Hugging Face forums, Reddit r/MachineLearning, r/LocalLLaMA, AI Twitter/X
 - Courses: fast.ai (free), deeplearning.ai short courses (free/paid), Hugging Face NLP course (free)
 - [deeplearning.ai](https://www.deeplearning.ai): <https://www.deeplearning.ai> | Hugging Face: <https://huggingface.co/learn>
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